

Decision Theory

Solution Sheet 5 and 6

Sheet 5: Hypothesis Testing and N-P Lemma

Problem 1: Loss Matrix in Hypothesis Testing

$H_0 : \theta = \theta_0, H_1 : \theta = \theta_1$. Risk: $R(\theta_0, \delta) = k_1 P_{\theta_0}(\delta = d_1)$ and $R(\theta_1, \delta) = k_0 P_{\theta_1}(\delta = d_0)$. Bayes Risk: $B(\pi, \delta) = p k_1 P_{\theta_0}(d_1) + (1 - p) k_0 P_{\theta_1}(d_0)$. This structure mirrors the Neyman-Pearson Lemma (Likelihood Ratio Test).

Problem 2: Poisson Test

$X_i \sim \text{Poi}(\lambda), n = 20$. $H_0 : \lambda = 0.1$ vs $H_1 : \lambda = 0.5$. $C = \{\sum x_i \geq 6\}$. Power function $\Pi(\lambda) = 1 - P(Y \leq 5)$ where $Y \sim \text{Poi}(20\lambda)$. Size $\alpha = \Pi(0.1) = 1 - \sum_{y=0}^5 \frac{e^{-2} 2^y}{y!} \approx 0.0166$. Power at H_1 : $\Pi(0.5) \approx 0.933$.

Sheet 6: UMP Tests and Exponential Family

Problem 1: Approximating Size with CLT

From Sheet 5, using CLT for $Y = \sum X_i$:

$$\alpha = P\left(\frac{\sum x_i - 20\lambda_0}{\sqrt{20\lambda_0}} \geq \frac{6 - 2}{\sqrt{2}}\right) = 1 - \Phi(2.828) \approx 0.0023$$

Problem 2: Gamma as Exponential Family

$$f(x; \alpha, \beta) = \exp((\alpha - 1) \ln x - \beta x + \alpha \ln \beta - \ln \Gamma(\alpha))$$

This matches the exponential family form $\exp(c(\theta)T(x) + d(\theta) + S(x))$, proving it has a Monotone Likelihood Ratio.

Take-Home Assignments

1. Admissibility of $\hat{\sigma}^2$: $X_i \sim N(\mu, \sigma^2), n = 5$. $\hat{\sigma}^2 = \frac{1}{5} \sum (x_i - \mu)^2$. $\text{Var}(\hat{\sigma}^2) = \frac{2\sigma^4}{5}$. Risk comparison shows whether estimators are R-better depending on bias and variance.

2. Bernoulli UMP Test ($n = 100$): $H_0 : p = 1/2$. By NP Lemma:

$$\frac{L_0}{L_1} = \left(\frac{p_0}{p_1}\right)^{\sum x_i} \left(\frac{1-p_0}{1-p_1}\right)^{n-\sum x_i} \leq k$$

If $p_1 > p_0$, critical region is $\sum x_i \geq k'$. If $p_1 < p_0$, region is $\sum x_i \leq k'$. (Two-tailed test context).

3. Uniform Distribution MLE: $f(x, \theta) = \frac{1}{2}\theta^{-1/2}x^{-1/2}$ for $0 < x < \theta$. Likelihood $L \propto \theta^{-n/2} I_{(x_{(n)}, \infty)}(\theta)$. Maximizing this requires minimizing θ , so $\hat{\theta}_{ML} = \max(x_i) = x_{(n)}$.